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Engineering Diploma in Telecommunication

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# Classification of human physical activity using a smart textile

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***Submitted by***


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# Supervisor Notes

# Abstract

Mobile health (m-health) covers medical and public health practices based on mobile devices such as mobile phones, patient monitoring systems, PDAs (personal digital assistants) and other wireless devices. It also includes lifestyle and wellness applications that can connect to medical devices or sensors as well as customized consulting systems.

Mobile health is a new field of activity whose rapid development can help to change health care and increase its quality and efficiency. This evolution is made possible by new smart devices, such as phones tablets and smart garments that contain sensors and that can collect several types of data.

One specific area of Mobile health is human monitoring that envolves human activity recognition (HAR). HAR is abroad field of study concerned with identifying the specific movement or action of a person based on sensor data.

This project is part of this line of research. The objective of this project is to present modestly a few steps along the road that will lead to the development of a classification system of human activity using machine learning and, more particularly, deep neural networks.

**Key words:** HAR, Human Activity recognition, Hexoskin, Wearable sensor, multimodal, machine learning, deep learning, context aware, combination of sensors, classify, Activity-transition detection

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# Dedication

*To my parents,*

*My reason for being, I miss the words to express all the gratitude and deep love that I have for the sacrifices you have made for my success and for your encouragements throughout my journey that I am at this stage of my life. I hope I have been worthy of your affection and confidence. May you find here the testimony of my gratitude, gratitude and respect, may God bless you with good health and long life.*

*To my brothers and sisters,*

*To whom I know that my success is very important, may God protect you and give you a life full of happiness and success.*

*To my dear friends,*

*In testimony of the sincere friendship that has bound us and the good times we spent together. I dedicate this work to you by wishing you a bright future and full of good promises.*

*To all those who are dear to me ...*

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# Chapter 1

# General Introduction

This chapter aims to fit the project in its general context, accordingly we will present the host organizations and the goal of the project to understand it well and define what is required to meet the specifications in a consistent and effective way.

## 1.1 Introduction

The last decade has revealed a rapid and significant growth of computing capacity driven mainly by decreasingly low cost of processors and tremendous progress in the nano-electronic architectures. Forthwith technology becomes undoubtedly ubiquitous in society permeating every aspect of daily life. In particular, data from different kind of sensors have marked a stupendous growth providing opportunities to use these data in ways that can make life changes. Correspondingly, multitudinous works have emerged in the purpose of exploring the theoretical and practical applications of technology especially in relation to human monitoring, treatment and healthcare.

As areas with huge potential application, human monitoring, treatment and healthcare are becoming requisite for a new era that witnesses the boom of data. Moreover, understanding our exercise habit, mental and physical health come to be required especially for elderly and people in rehabilitation to help to monitor and improve their health and well-being. Rebecca Hussey, physiologist at AXA PPP healthcare [1], says: *"Under-*

[1] AXA PPP healthcare, originally known as The London Association for Hospital Services, is a UK private medical insurance provider and was set up in 1938 as a private healthcare scheme for people of middle income in London.[healthcare, 2019a]

*standing our health and tracking our progress has become much easier now that we're able to use data collected on a daily basis from health tech devices. From fitness bands to smart watches and sensors, and tech with medical applications like blood glucose monitors, health tech is now being widely adopted by many people who want to take personal responsibility for their health."*

One specific area of human monitoring is human activity recognition which is a broad field of study concerned with identifying the specific movement or action of a person based on sensor data. Historically, sensor data for activity recognition was challenging and expensive to collect, requiring custom hardware. But nowadays it becomes easier with smart-phones and wearable sensors ranging from smart wristbands devices, which appear to be an adhesive bandage but are in fact skin-worn circuits, to smart shoes. These sensors allow us to collect a large amount of data that need to be analyzed using data mining and machine learning techniques. Conventionally, machine learning techniques, which rely on hand-crafted features, are used in developing Human Activity Recognition models. These techniques require feature engineering and expertise in data analysis and signal processing, making them expensive in term of human resources and engineering time. However recent developments in Deep Learning have greatly advanced the performance on many tasks such as Human Activity Recognition, promising for auto-learning high-level features from raw data, to say nothing of its tendency to grow exponentially with data quantity. Thus, it is necessary to emphasis the fact that the growth in deep learning performance does not mean that the era of machine learning has ended. Correspondingly Alfred Essa[2] says: *"Traditional statistical learning does not disappear. It solves a lot of problems, Most of the Machine Learning we do is not Deep Learning. We work with relatively small datasets. Various machine learning techniques can be tried and applied in a relevant way. That's what the good Data Scientists do."*. Furthermore, researchers performed feature engineering to extract effective handcrafted features to decrease the computational cost while maintaining good accuracy, unlike deep learning techniques that are "greedy" in terms of computational cost. However, in most cases, the improvement of accuracy is at the expense of increasing the computational cost [Jian Cui, 2013]. With this intention,

[2]Alfred Essa: Vice President of Research and Data at McGraw-Hill Education, a New York-based publishing house that is a world leader in academic and scientific publishing.

the goal of this work is to produce a novel model for Human Activity Recognition through a deep learning approach. Equally important, the project aims to develop another model based on traditional machine learning techniques for the purpose of comparing the two approaches.

## 1.2 Project context

This project is included as a mandatory part in the training of engineering students at the higher school of communications of Tunis (Sup'Com, Tunis, Tunisia) to enhance their scientific and technical abilities. It is held by the collaboration of TELUQ university[3] and Cr-CHUM[4] research center since they are working jointly in various themes particularly artificial intelligence and e-health. Precisely, this project is affiliated to the research center LICEF[5] and to the Cr-CHUM's research axis LIO [6] in partnership with the Canadian company *Carré technologie Inc.* as a research project lasting five years from January 2015 for the development of a classification system for human activity recognition that will be used later in the development of a system to measure the compliance of patients in home rehabilitation. Hence, this work falls in the Context-Aware Sensing field based on body-worn sensors for the purpose of creating a classification system for Human Activity Recognition (HAR) to be used in patient monitoring and care.
Correspondingly, the Hexoskin is used as a "smart" textile designed by the *"Carré Technologie Inc."*. The Hexoskin Smart Shirts are clinically validated to continuously track heartbeats (ECG), lungs, sleep, and activity data and are equipped with three sensors:

- One accelerometer that measures three dimentional kinetic data of the person. (Figure 1.1)
- Three electrodes embedded in the textile to measure the electrocardiogram (ECG)

[3]TELUQ: originally Telé-université, Université du Québec, is a part of the Université du Québec system offering french-thought courses via distance learning.

[4]*Cr-CHUM: Centre de recherche du Centre Hospitalier de l'Université de Montréal* translated as The research Center of the University of Montreal Hospital Centre, is is one of the biggest and most major of the health-care, clinical and scientific research centers in north America.

[5]originally *LICEF: Laboratoire en Informatique Cognitive et Environnement de Formation* translated as Laboratory in Cognitive Computing and Training Environment.

[6]LIO: originally *Laboratoire de recherche en imagerie et orthopédie* translated as the Imaging and Orthopedic Research Laboratory.

signal. (Figure 1.2)

- Two bands integrated in the clothing and operated by inductance plethysmography[7] that records respiratory signals at the thoracic and abdominal levels. (Figure 1.3)



Figure 1.1: Hexoskin - Accelerometer



Figure 1.2: Hexoskin - ECG sensors



Figure 1.3: Hexoskin - Respiration sensors

Accordingly, the work presented in this report is the continuation of a research program [Nour Cherif, 2018b] initiated in the (LIO), the Centre de Recherche LICEF (TELUQ university , Montreal, Canada), and COSIM Lab Sup'Com, (Tunis, Tunisia). In this context, a study explored human activity classification using conventional machine learning techniques. Hence we are in this project, we are aiming to test deep learning techniques in HAR problem compared to machine learning techniques.

## 1.3 Project presentation

The recognition of human actions is a classification problem of data provided by a specialized textile or by the smartphones sensors. Since 1980, it has become a very important and challenging research field due to its connection to various study fields and to the variety of applications that offers in correlation with several domains (such as health, intelligent environment, monitoring and security). From this fact, the objective of this research project is to classify physical activity based on kinetic data and physiological data provided by a "Hexoskin" connected textile.

[7]Respiratory inductance plethysmography (RIP) is a method of evaluating pulmonary ventilation by measuring the movement of the chest and abdominal wall.

To better understand all of the dimensions of the subject and give an almost exhaustive description of the situation, we followed the questioning method WWWWHW (What, Who, Where, When, How, Why):

- **What:** This research project falls under the topic of human physical activities classification (HAR: human Activity Recognition) which is the subject of several scientific and academic researches.
- **Who:** The solution must be able to classify the activities of any human being regardless of their characteristics such as their age, size, weight.
- **Where:** The classification must be possible in any place, for instance at home and at work.
- **When:** The classification is performed on daily activities such us running, jumping and falling.
- **How:** To do this, we are using the machine learning tools for the creation of the model and a Hexoskin smart fabric (having an acceleration sensor, an ECG signal sensor and a respiratory signal sensor) for the data acquisition.
- **Why**: The goal is to find a powerful model for the classification of human activities that is based on a multimodality of heterogeneous sensors: kinetic sensors (movement) and physiological sensors.

## 1.4 Adopted formalism : ISO 13485: 2016

The LIO has a Quality Policy that provides a framework for the realization of its projects: ISO 13485:2016.

ISO 13485:2016 Medical devices – Quality management systems – Requirements for regulatory purposes is an International Organization for Standardization (ISO), specifies requirements for a quality management system where an organization needs to demonstrate its ability to provide medical devices and related services that consistently meet customer and applicable regulatory requirements. Such organizations can be involved in one or more stages of the life-cycle, including design and development, production, storage and distribution, installation, or servicing of a medical device and design and development or provision of associated activities (e.g. technical support). ISO 13485:2016 can also be used by suppliers or external parties that provide product, including quality management

system-related services to such organizations. [https://www.iso.org/standard/59752.html]




Figure 1.4: ISO 13485:2016

LIO aims to provide a vibrant and enriching training venue for sharing and learning science along with increasing the credibility and recognition of the LIO by demonstrating its competences, its professionalism, its innovative research and its perpetual evolution in the field of technology transfer..

## Conclusion

Throughout this chapter, we have set the project in its general framework, namely the presentation of the project and its objectives. The rest of the report is structured around four chapters.

In the second chapter ***State of the Art***, we draw up the state of the art for human activity recognition describing different architectures and clarifying their limits. A particular attention will be given to related works through a Deep learning approach. Next we will be moving to the fourth chapter

Next we move to the third chapter ***Methods and Experimental Set-up***. In this chapter, the experiment setting will be explained step by step starting from the prepossessing stage up to the fine-tuning phase.

In the fourth chapter ***Result and Analysis***, the results of the experiment are analyzed and interpreted .

In the last chapter ***Conclusion and Future Work***, the most important results are summarized and synthesized, limits of this work are presented and some indications are stated for future improvement.

# Chapter 2

# State of the Art

Human activity recognition is q problem of classifying sequences of data recorded by specialized harness or smart phones into well-known movements or activities. This is a challenging problem given the large number of observations produced each second, their temporal nature and the lack of a clear analytical way to relate the sensor data to specific actions in a general way. This chapter aims to provide a brief presentation of previous work related to human activity recognition problem. It contains a description of available datasets , the metrics and different validation protocols that are used in litterature.

## 2.1 Datasets for human activity recognition problem

### 2.1.1 Available datsets

The most used datasets in HAR problem are reported in the following paragraphs. Most of them are open-source database for research purposes.

**MHEALTH**: The primary goal of this dataset is to enable the development and validation of mobile health applications. For this purpose, MHEALTH dataset includes records of body movements and vital signs of ten volunteers of different profiles, while performing several physical activities [O. Banos and Villalonga, 2014]. These signals are recorded at a sampling rate of 50Hz using several sensors placed on the subject's chest, right wrist and left ankle to measure multiple signals acceleration, rate of turn, magnetic field orientation and electrocardiogram signal. However its compulsory to mention that the latter signal is not used in multiple studies since it contains a large number of damaged samples.

**PAMAP2**: This dataset contains two groups PAMAP2-Protocol and PAMAP2-Optional [Reiss and Stricker, 2012]. Each group differs in the types of the performed activities. It also contains data of 18 different physical activities (such as walking, cycling, playing soccer, etc.), performed by 9 subjects wearing 3 inertial measurement units and a heart rate monitor. These sensors capture data using a sampling rate of 100Hz.

**USCHAD**: This dataset contains data for this dataset were captured using a wearable accelerometer sensor with a sampling rate of 100Hz. It currently includes 14 subjects and 12 daily activities. Moreover, it is a suitable dataset to CNN approaches due to their large number of samples and activities.

**UTD-MHAD (UTD)**: This dataset consists of a significant variability of activities captured using multimodal sensors including RGB videos, depth videos, skeleton positions, and inertial signals from a Kinect camera and a wearable inertial sensor [C. Chen and Kehtarnavaz, 2015]. UTD-MHAD was divided into two groups of activities according to where the inertial sensors are placed. In the first group (UTD-MHAD1) the inertial sensors were placed on the subject's right wrist, while the second group (UTD-MHAD2) the sensors were placed on the subject's right thigh.

**WHARF** is the most recent dataset to perform human activity recognition based on wearable sensors :[Bruno et al., 2015]. Activities are captured using an accelerometer sensor with a sampling rate of 32Hz. Moreover, this dataset is composed of 14 activities;

**WISDM**: The WISDM [healthcare, 2019b] dataset contains the largest number of samples compared to the other datasets . It present few categories of activities recorded with a small sampling rate and from a high number of subjects.

### 2.1.2 Used sensors for human activity recognition

Throughout literature, the datasets were collected using simple sensors namely accelerometers (Acc), gyroscopes (Gyro), magnetometers (Mag), temperature (Temp) sensors. However these sensors used different sampling rates due the various modalities of sensors from different manufacturers. Also, it is necessary to mention that having low sampling rate would reduce the computational cost. However this may have an impact on the prediction accuracy [Krause et al., 2005]. Consequently a trade-off between the sampling rate and the prediction accuracy is required. The following table 2.1 illustrates the different sensors used by the state-of-art datasets mentioned above.

| **Dataset** | **Frenquency (Hz)** | **Sensors** |
|---|---|---|
| MHEALTH | 50 | 3 (Acc, Gyro, Mag) |
| PAMAP2 | 100 | 4 (Acc, Gyro, Mag, Temp) |
| USCHAD | 100 | 2 (Acc and Gyro) |
| UTD-MHAD1 | 50 | 2 (Acc, Gyro) |
| UTD-MHAD2 | 50 | 2 (Acc, Gyro) |
| WHARF | 32 | 1 (Acc) |
| WISDM | 20 | 1 (Acc) |

Table 2.1: Different datsets and their corresponding sensors. Acc, Gyro, Mag and Temp indicate respectively accelerometers, gyroscopes, magnetometers, temperature sensor.

## 2.2 Related work

Several studies have addressed the human activity recognition (HAR) using conventional machine learning techniques and deep learning techniques which are discussed in the following two sections. The following table 2.1 summarizes the used techniques used in some related works.

| WORK | TECHNIQUE | DATA TYPES | DATA GENERATION | METRICS | VALIDATION PROT. |
|---|---|---|---|---|---|
| Multiscale DCNN Ensemble Applied to Human Activity Recognition Based on Wearable Sensors [Jessica Sena, 2018] | Multimodal DCNN ensemble (using SELU activation function & 2D ConvLayers) | accelerometer, gyroscope, magnetometer and temperature | Sliding Temporal Window (unknown overlapping status) | Accuracy | Leave-One-Subjet-Out & Leave-One-Trial-Out |
| Context aware adaptable approach for fall detection bases on Smart textile, [N. Mezghani, 2017] | Fall detection & orientation using SVM classification (& downward zero-crossing of the smoothed first derivative of signal A for peak detection) | accelerometer | region of interest after peak detection | Accuracy | 10-fold cross validation |
| Wearable-Based Human Activity Recognition Using and IoT Approache, [Diego Castro, 2015] | DCT + PCA + structural detector + classifier (Bayes & Decision tree C4.5) | accelerometer, ecg magnetude, respiration rate, posture, peak acceleration, heart rate | Sliding Temporal Window (unknown overlapping status) | Precision | – |
| DeepSense A Unied Deep Learning Framework for Time-Series Mobile Sensing Data Processing, [Shuochao Yao, 2015] | 3 Individual Convolutional Layers + 3 Merge Convolutional Layers + 2 Recurrent Layers (GRU) | motion sensors (accelerometer, gyroscope) | series of non-overlapping time interval | accuracy, macro F1-score, and micro F1-score | Leave-One-Subjet-Out |
| Latent HyperNet Exploring the Layers of Convolutional Neural Networks, [Artur Jordao, 2018] | Latent HyperNet (output of each convLayer is used for classification ) +PCA for dementionality reduction | accelerometer, gyroscope, magnetometer | Sliding temporal Window (– overlapping status). The windows that do not t within the temporal window is dropped | Recall | 10-fold cross validation |
| A Deep Learning Approach to Human Activity Recognition Based on Single Accelerometer, [Yuqing Chen, 2015] | CNN (2D kernel) | single accelerometer | Semi-Non-Overlapping Window | accuracy | Train-Test split |
| Multi-Modal Convolutional Neural Networks for Activity Recognition, [Sojeong Ha, 2015] | Multi-Modal CNN (2D kernel) | accelerometer, gyroscope, magnetometer | Semi-Non-Overlapping Window | accuracy | Train-Test split |
| Convolutional Neural Networks for Human Activity Recognition using Multiple Accelerometer and Gyroscope Sensors, [Sojeong Ha, 2016] | Multi-Modal CNN (2D kernel) (CNN-pf partial weight sharing & CNN-pff partial and full weight sharing in the 1st convLayer) | accelerometer, yroscope, magnetometer, ecg | Semi-Non-Overlapping Window | accuracy | – |
| HumanActivity Recognition using Wearable Sensors by Deep Convolutional Neural Networks, [Wenchao Jiang, 2015] | Multi-Modal DCNN model (2D kernel) & Multi-Modal DCNN+ model (using SVM to discriminates in case of uncertainty) on Signal image | accelerometer, GYROSCOPE | Sliding temporal Window (– overlapping status). | accuracy | – |



Table 2.2: Techniques, data generation, metrics and validation protocol employed by different works

### 2.2.1 Human activity recognition using methods based on Handcrafted Features

To represent the activities, [N. Mezghani, 2017] and [Nour Cherif, 2018b] extracted handcrafted features from the raw signal starting from time-domain features (i.e.minimum, maximum, mean, variance (VAR), standard deviation (STD) etc...), to frequency domain features (i.e.,maximum magnitude of the Fast Fourier Transform (FFT)). These features are computed on region of interest (ROI) defined around detected peaks. The authors also analyzed the ability of SVM to separate the categories of activities. To decrease the unwanted behavior of the signal,the author in [Diego Castro, 2015] used principal component analysis (PCA) or a discrete cosine transform (DCT) to extract the most relevant information of the signal.

### 2.2.2 Human activity recognition using deep learning techniques

One of the most comprehensive studies in human activity recognition based on wearable sensors is the works of [Yuqing Chen, 2015], [Artur Jordao, 2018], [Jessica Sena, 2018], [Sojeong Ha, 2015], [Sojeong Ha, 2016] and [Wenchao Jiang, 2015]. The described works use convolutional neural networks (ConvNet) to learn characteristics and classify them simultaneously. Focusing on convolutional neural networks, Chen and Xue employed a sophisticated ConvNet, where the input is taken from the raw signal and is passed to three blocks of convolutional layers followed by a maxpooling layer. Another relevant work is the research held by [Jessica Sena, 2018] where they proposed a ConvNet architecture consisting of three multiple deep CNNs (DCNN), with the intention of fusing later the output of all DCNNs. They employ an ensemble of DCNNs with different kernel sizes each to extract information from multiple temporal scales. They trained their model on the available open source datasets (mentioned above) and using Leave One Subject Out (LOSO) cross-validation protocol. Assuming that different detection sensors (ie accelerometer, gyroscope and magnetometer) should be used, [Sojeong Ha, 2015] introduced a zero-padding between each heterogeneous modality of ensor to prevent them from being merged during the convolution process. In another study [Artur Jordao, 2018], the author proposed a different concept to exploit CNNs architectures. He introduced the multiscale learning by exploiting the output of each layer for classification (after dimentionality re-

duction using PCA). The proposed is trained model with some state-of-the-art datasests using the 10-fold cross-validation protocol.

# Chapter 3

# Methods and Experimental Set-up

Having introduced the state-of-the-art techniques used in HAR problem, in this chapter we are presenting our proposed methods for HAR problem starting from the establishment of the dataset using the Hexoskin, to the evaluation of our model as described in figure 3.1.

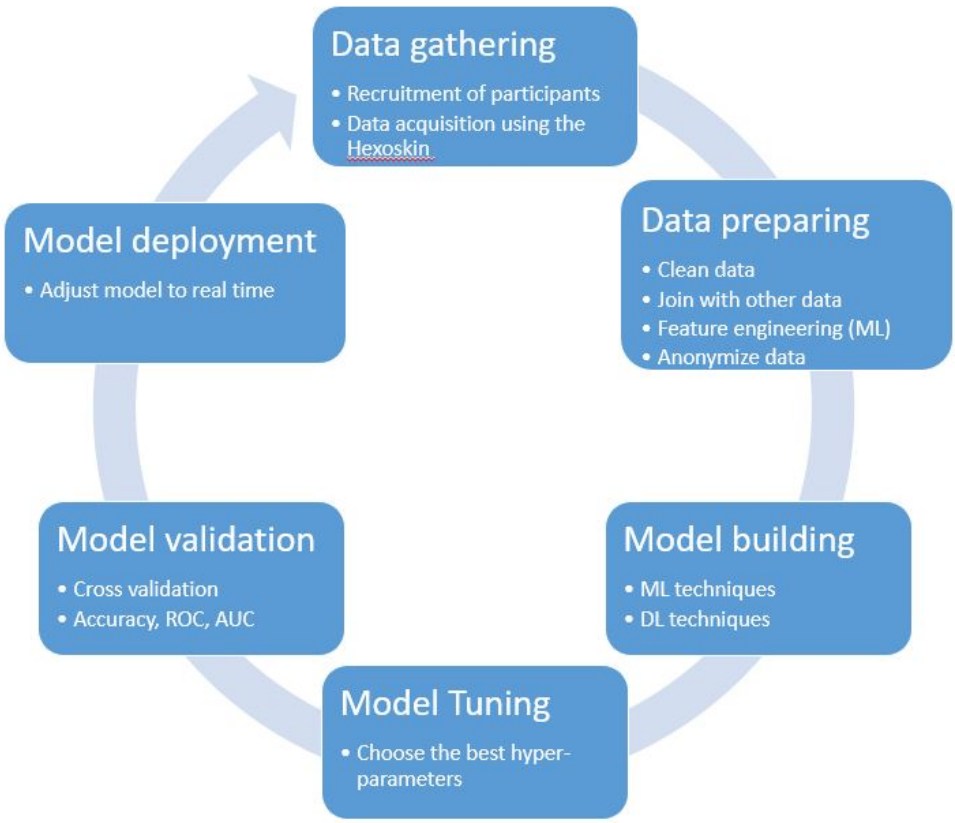



Figure 3.1: Project workflow

## 3.1 Data collection

### 3.1.1 Ethics

This project is submitted to the research ethics committees of the CHUM, the École de technologie supérieure and the Téluq. The fundamental ethical principles as well as the guidelines of the councils are respected during the project. All recruited participants participated voluntarily knowing that they might withdraw at any time. The nature of the assessments, their progress and the risks involved in participating in this study were explained to the participants when signing the consent form. All information collected during this study are remaining strictly confidential, within the limits provided by Canadian law. A code were assigned to each participants throughout the study. The list of participants and their personal details will remain confidential at the LIO for a period of 5 years. Only authorized persons (principal investigator and student researcher) will have access to it. Then, personal information about the participants as well as any means of identifying them are destroyed. The data obtained may be the subject of one (or more) scientific publication(s) but may not lead to any dissemination of the information of participants in this study.

### 3.1.2 Participants

For this work, a total of 40 people were recruited internally, at the Imaging and Orthopedics Research Laboratory (LIO) of the École de technologie supérieure (ÉTS) located at the CHUM Research Center, to participate in this study.
Participants were recruited from the laboratory members at CrCHUM. Participants were verbally informed that topics are being sought for acquisitions and that those interested in participating would be able to mention it to the research professional. No pressure were exerted on the members of the laboratory to participate in the study, their participation were completely voluntary. Recruited participants must be in good health and should not have chronic pain or known locomotor or heart problems. Pregnant women were not to be recruited for this study. If a participant were or suspects they are pregnant, they would not be able to participate. No pregnancy test were done, it would be verbally asked to the participant if she is pregnant.

### 3.1.3 Protocol

All participants wore a Hexoskin. Using the Hexoskin device, several data such as the three axis acceleration signal, the electrocardiogram signal, etc..., are collected The different data provided by these sensors are recorded on a data server at the LIO.

#### The acquisition procedure steps

Once a person indicates that he is available, the room of the biomechanical laboratory is reserved and an information and consent form (ICF) is sent to the participant. The latter must read the ICF, sign it and bring it with him on the day of the acquisition.
The day of the acquisition, a 5-minute period were given to the participant to show them how to wear the Hexoskin and to allow them to get used to the realization of various physical activities. This adaptation period would also allow us to determine if the participant is comfortable with the different physical activities and to answer their questions.
Then each participant were asked to repeat twice three cycles containing the following fifteen physical activity classes:

- climbing stairs (C1) on 2 floors
- go down stairs (C2) on 2 floors
- walk (C3) for 30 seconds
- run (C4) for 30 seconds
- sit down (C5) for 30 seconds
- crouching (C6) for 30 seconds
- fall on the right side (C7) on a padded air mattress
- fall on the left side (C8) on a padded air mattress
- fall forward (C9) on a padded air mattress
- fall back (C10) on a padded air mattress
- lie on the right side (C11)
- lie on the left side (C12)
- lie on the stomach (C13)
- lie on the back (C14)
- squat five times (C15)

=For all activities requiring a fall, namely, class C7, C8, C9 and C10 the research team showed the participant in advance what is the safest way to perform the activity. The fifteen activities were carried out according to three different cycles (depending on the order of the activities) to obtain a more realistic representation of the different possible transitions between the fifteen classes. A period of 5 minutes of rest between each series is also envisaged. Throughout these steps we have collected the data. A video recording were made during the whole session. The recorded video sequences allow us later to label the different classes of physical activities.

#### Collected Data

The data collected are as follows:

- Anthropometric data of participants (weight, height, sex, etc.);
- Accelerometer, gyroscope, speed, gravity data, etc. are recorded from the sensors.
- Cardiac activity data (heart rate, heart rate variability HRV (to estimate stress and fatigue), physical recovery assessed from changes in heart rate, etc.) are measured by Hexoskin.
- Respiratory activity data (respiration rate (RPM) and minute ventilation[1]) are also be measured by Hexoskin.
- Video cameras record video clips to label different classes of physical activity (walking, running, sleeping, falling, climbing and descending stairs). These labels will serve as a reference for the validation of data analysis and classification methods. All videos are denominated, faces are blurred at the end of the session using third-party software.

### 3.1.4 Hexoskin system

#### System description

The Hexoskin is composed of a garment that incorporates the heart sensors and respiratory [Fournier and Roy, 2011]. The set of sensors is connected to a special connector to ensure

[1]Minute ventilation is the volume of gas inhaled (inhaled minute volume) or exhaled (exhaled minute volume) from a person's lungs per minute.

the connection to the Hexoskin device. The module contains the electronics necessary for the acquisition, recording and data transmission. It also contains the 3-axis accelerometer. Table 3.1 presents the product components and their characteristics.

| **Smart textile** | **Hexoskin module** | **USB Cable** |
|---|---|---|
|  |   |  |
| Italian high fabric<br>performance<br>Machine washable<br>Dry quickly<br>Good breathing<br>Anti-odor and UV protection | Battery life +14 h<br>Continuous recording over<br>+150 h<br>Bluetooth connection | Hexoskin / USB connector<br>for data transfer<br>and to recharge the module |

Table 3.1: Hexoskin components

To use the product, the first step is to assign a user to the Hexoskin device. The module must be connected using the USB cable to the computer. The association of the name with the module is carried out with the HxService software. One can also check the state of charge of the battery. To start recording, simply connect the device to the connector of the garment. Following recording step, we disconnect the device from the garment and connect it to a computer with an Internet connection. The data is then sent to the Hexoskin server. Then, after obtaining the consent of the participants, they are invited to put on their sport outfit. That's when the Hexoskin garment needs to be put on. We make sure we have the best fit possible. Thus, to have good signals,

the garment should be tightened as much as possible while remaining comfortable. It is also recommended that the garment should be placed with care to avoid breaking the connections. Subsequently, when the participant starts performing the different activities, the Hexoskin device is plugged in for the start of the recording. The module is then placed in the pocket, always in the same orientation for purposes of uniformity. The device is disconnected from the garment at the end of each trial. At the end of all trials, the participant can remove the garment.

### Recorded signals

The Hexoskin product can record 3 types of signals, ie cardiac, respiratory and movement. Table 3.2 describes each of the channels in more detail.

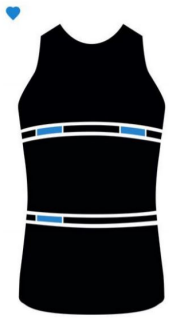
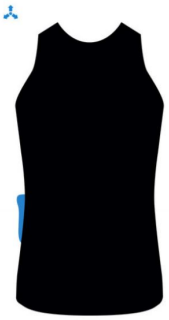
| Cardiac | Respiratory | Movement |
|---|---|---|
|  |  |  |
| - 1 ECG lead<br>- 3 soft cloth electrodes driver: 2 at the thoracic level for ECG and 1 at the abdominal level serving grounding | - 2 belts integrated in garment operated by plethysmography by inductance:<br>- 1 thoracic<br>- 1 abdominal | - 1 integrated 3-axis accelerometer in the Hexoskin module.<br>The garment contains a pocket to put the module |
| Gross cardiac signal, detection of QRS peaks, RR interval, rhythm cardiac | 2 gross respiratory signals, breath detection, respiratory rhythm, ventilation minute, tidal volume | Gross acceleration signal: 3 axes, intensity measurement activity, cadence and step detection |
| Analog signal<br>Sampling frequency 256 Hz | Analog signal<br>Sampling frequency 128 Hz | Analog signal<br>Sampling frequency 64 Hz |

Table 3.2: Product components

### 3.1.5 Data labelling

#### Protocol

One mandatory part to elaborate database for HAR is the process of labeling the recorded data hence we are facing a supervised learning problem. To put it differently Labeling is still seriously business given that each mistake or inaccuracy negatively affects a dataset's quality and the overall performance of a predictive model. Under those circumstances, we operated as follow:

* First, we organized all participants files ( data + video ) following a well-defined tree in the data-server as shown in the figure 3.2.

* Second, to get a high-quality dataset, we labeled our data as finely as possible. Consequently, we recorded the activities performed by all participant with the time shown in the video in milliseconds as shown in the figure 3.3. Also, it has been ensured that the time shown in the video is synchronized continuously via internet.

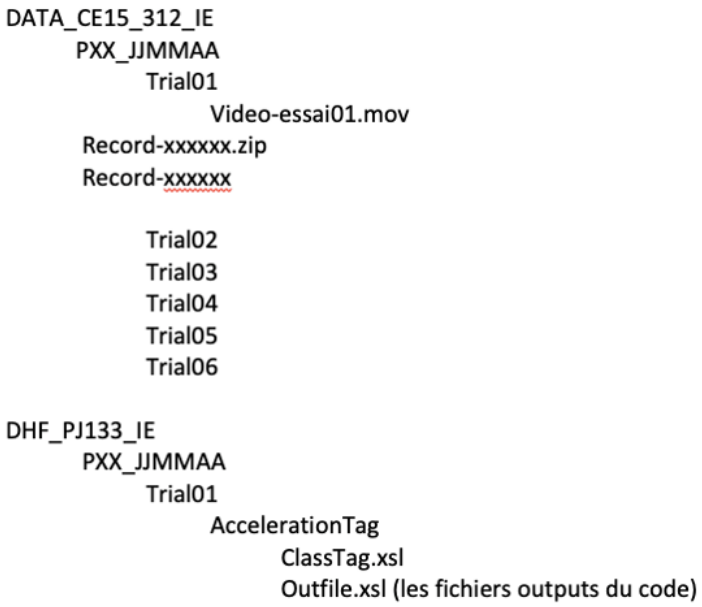



Figure 3.2: Data tree structure in the data-server.




Figure 3.3: Video recording labeled with real time.

* Then for each trial, we loaded the video and we annotated the start and the end of each activity using the time shown in the video. Similarly, we annotated all the transition activities. All the annotated start  end of activities of each trial are saved in excel file with a generic name "ClassTag".

* At the same time, we generate all participants files containing all the acceleration signal annotated by the time provided by the hexoskin box.

* After finishing the labeling of all participants' data, we move to the process of generating files that contain the labeled acceleration data using the annotated start  end of each activity. Now that the labeled data is ready, we validate the process by plotting the generated data using a different color for each activity as shown in the figure 3.4.

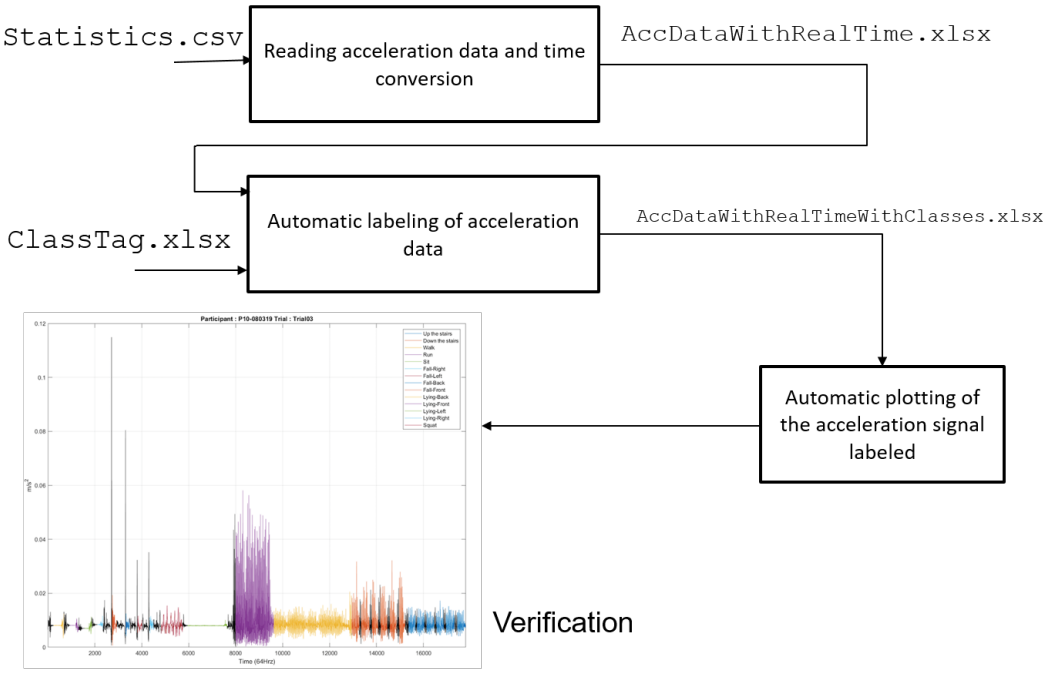



Figure 3.4: Acceleration signal labeling.

### Encountered problem

At the labeling level, there was a small temporary gap between the time provided by hexoskin and the time provided by the mobile application with which video capture was done. To fix this problem, we manually estimated the temporary offset based on the generated graphs (containing the acceleration data presented with different color for each activity).

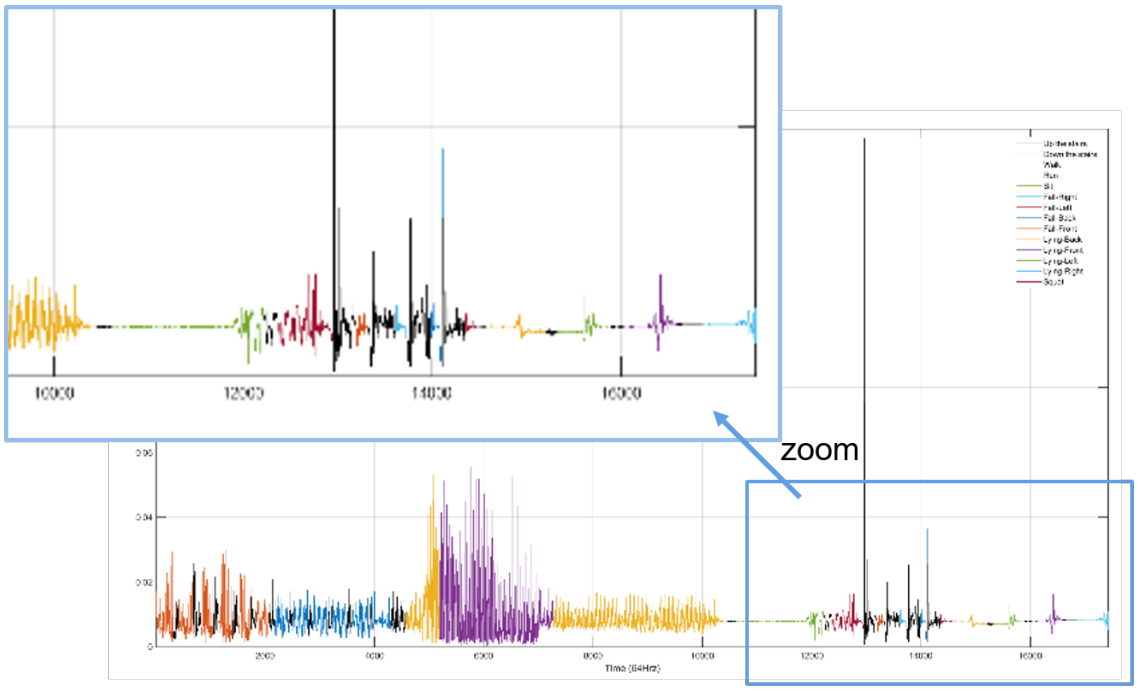



Figure 3.5: Temporary gap in data labeling

## 3.2 Proposed methods

### 3.2.1 Machine learning approach

#### Architectural design

Using the conventional machine learning techniques, our method follows the block diagram of figure 3.6.

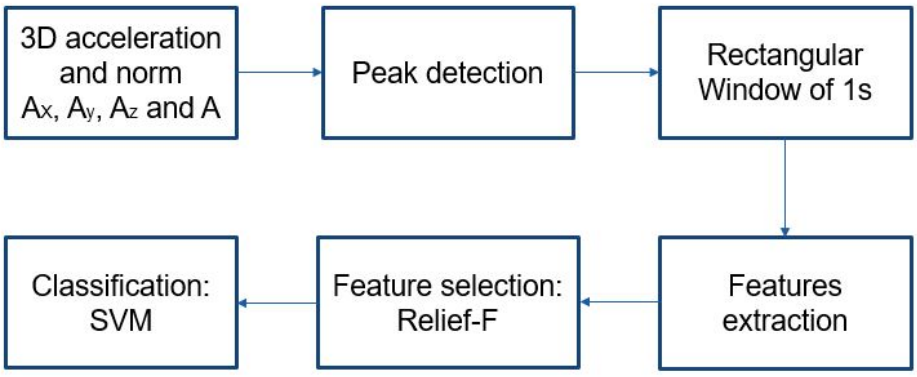



Figure 3.6: Classification System based on SVM

The goal of the SVM is to find a separator between two classes that is at most away from any point of the training data. If we come to find a linear separator that is to say that there is a separating hyperplane then the problem is said linearly separable otherwise it is not linearly separable and it there is no separating hyperplane.

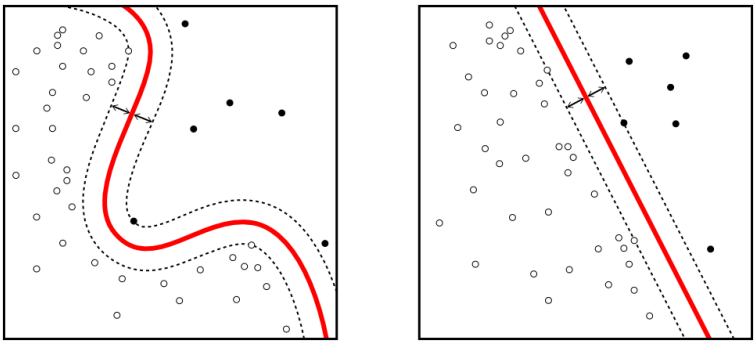


Figure 3.7: Acceleration signal labeling.

SVMs have several advantages. Indeed, they have a solid theoretical foundation, they SVMs are effective in large spaces and they accept different kernel functions.
However, despite their performance, the SVM presnt some limitations. Indeed they use complex mathematical functions for classification and require a huge amount of time during the test phases.

### Peak detection and windowing

Let Ax, Ay and, Az the accelerations along respectively the x, y and, z axes. The norm A of the acceleration vector is defined by:

$$A = \sqrt{A_x^2 + A_y^2 + A_z^2} \tag{3.1}$$

Firstly, the norm of the acceleration is computed from the three axis of the acceleration signal. The sequence of the norm of the acceleration vector is secondly segmented into short time sequences using a rectangular window around each detected peak in the sequence. More precisely, peaks are detected by thresholding the norm. The value of the threshold was adjusted in a previous work to 1.1 [Nour Cherif, 2018a]. According to this work, the size of the rectangular window was also fixed 64 point around each detected peak so as its temporal duration is around 1 second as recommended in [Banos O, 2014] where a temporal window between 1 second and 2 seconds proves to provide the best trade-off between recognition speed and accuracy. Another reason to take a window size

of 1 second is the fact in the case of small window inferior of 1 second, we may loose important information in the signal as we will be dealing with small amount of samples. Moreover, a bigger window may be bigger than the time required to an activity to be finished as the case of falls which lasts between 1 seconds and 2 seconds. For each window, several features are computed to characterize the signal as described in the following paragraph.

**Feature extraction**

For each window, several features are computed on the 4 signals Ax, Ay, Az and the norm A. We recall that features resumes the proprieties of the signal into a set of descriptors [Pyle, 1999].

For the sake of simplicity, let Sn denote either of the component Ax,Ay, Az or A at time n. For each Sn, the following time-domain features have been considered in each window, minimum, maximum, mean, variance (VAR), standard deviation (STD), kurtosis, skewness, energy, Root Mean Square (RMS), Zero Crossing Rate (ZCR), Euclidean norm, Maximum Absolute Column Sum (1-Norm), amplitude and correlation. The frequency domain feature used is the maximum magnitude of the Fast Fourier Transform (FFT). Also, some conventional statistical features are used such as the simple correlation value (Cor) between the three axis of the acceleration signal.

**Relief F**

The Relief-F algorithm was used to identify the most important features by ranking them according to a computed relevance score [I. Kononenko, 1997]. It is one of the popular filters based feature selection methods to estimate the weight of features. It can efficiently estimate the quality of features in classification problems with strong dependencies between features. It randomly selects an instance Ri then finds the k nearest neighbors from the same classes, called nearest hits Hj and the k nearest neighbors from the different classes, called nearest misses Mj(C). It updates the quality estimate W[A] for all features A according to their values for Ri , Hj and Mj(C). Instead of finding a single miss from a different class, Relief-F finds one near miss Mj(C) for each different class C and averages their contribution for updating the estimate W[A]. The average is weighted with the prior probability of each class P(C). The process is repeated for m times. User-

defined parameter k controls the locality of the estimates. $diff(A, I1, I2)$ calculates the difference between the values of the attribute A for two instances I1 and I2. Function diff is also used for calculating the distance between instances to find the nearest neighbors. The whole distance is simply the sum of differences over all attributes. A more detailed discussion can be found in [M. Robnik-Sikonja, 2003].

**Algorithm 1:** Algorithm Relief-F

**Input :** For each training instance, a vector of attribute values and the class value

**Output:** the vector W of the estimate qualities of attributes

```
1  Shuffle the dataset randomly.
2  Split the dataset into k groups Set all weights: W[A] := 0;
3  for i := 1 to m do
4      Randomly select an instance R_i;
5      Find k nearest hits H_j
6      for each class C 6= class(R_i) do
7          from class C find k nearest misses M_j(C);
8          for A := 1 to number ofattributes do
9              W[A] := W[A] - sum_{j=1}^{k}(diff(A,R_i,H_j)/(m×k)) + sum_{C6=class(R_i)}(P(C)/(1−P(Class(R_i)))) +
               sum_{j=1}^{k}(diff(A,R_i,M_j(C))/(m×k))
10         end
11     end
12 end
```

### Classification

After the selection process, we retained the top ten features to be used for the classification. In this work we investigated one traditional classification method: A support vector machine (SVM) which is a supervised machine learning algorithm that aims to find the optimal separating decision hyperplanes between classes with the maximum margin between patterns of each class. The classifier performances are evaluated in terms of classification accuracy and classification accuracies per class. For this purpose, we are using three different validation protocols, the k-fold, the Leave One Subject Out and the nested

cross-validation protocols which will be described latter.

### 3.2.2 Deep learning approach

#### Deep learning advantages

Deep learning techniques have several advantages as described in [Donges, 2019] :

1. They get better the more data you fed into them.

2. They does not require a feature engineering step as they learn high-level features from data in an incremental manner.

3. They can be used when there is lack of domain understanding for feature introspection.

#### Deep learning advantages

1. They are more computationally expensive than traditional algorithms

2. Their "black box" nature, meaning that you don't know how and why your algorithm came up with a certain output.

#### Architectural design




Figure 3.8: Classification System based on deep learning techniques

While the participants performed physical activities according to the above protocol (data acquisition protocol),the Hexoskin they wore measured acceleration signals that were then digitized and stored in the Hexoskin device. Acceleration signal acquisition was performed at a frequency of 64 Hz. Stored data files containing all activities were then downloaded onto a PC for data processing.

### Data preprocessing: segmentation

The major advantage of the models of the deep learning is the fact that it operates directly on the raw data, because of this, we just segment the data by means of a sliding window of size fixed between 1 second and 2 seconds (as justified above) and with an overlap of 50 percent between consecutive windows (as in Fig. 3.9B). In this work, we fixed the size of the sliding window to 1 second.

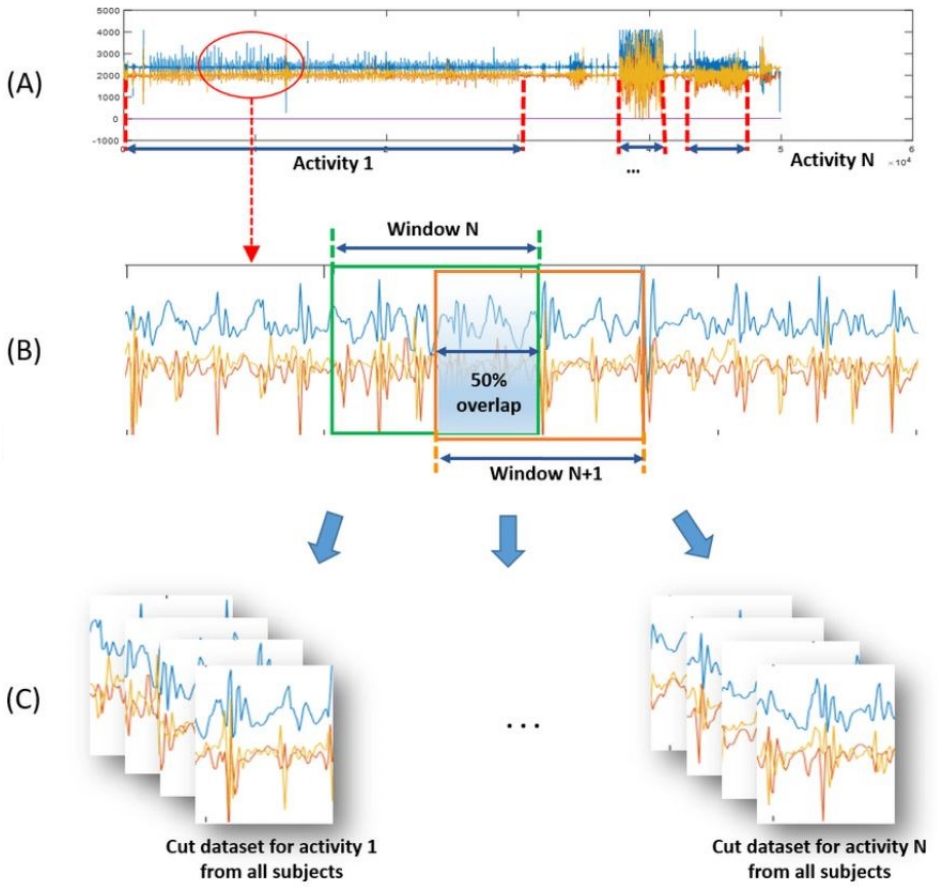



Figure 3.9: Acceleration signal segmentation using a sliding window. (A) Acceleration signal data file of each trial. (B) Acceleration signals segmentation and windowing description (Window Size = 64 points = 1s). (C) Dataset of activities from all subjects. [Jang et al., 2018]

### Proposed architecture

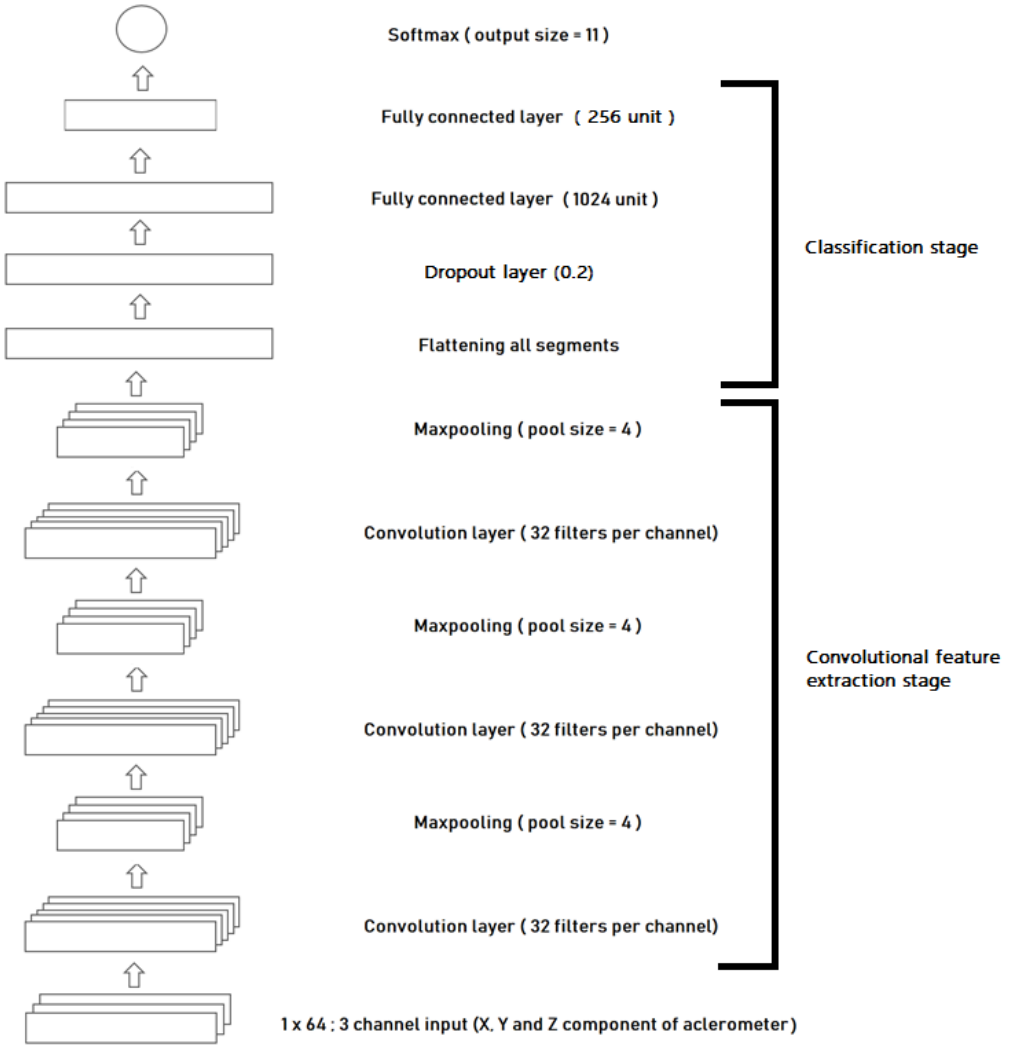



Figure 3.10: Convolutional neural network (CNN) structure with input, feature extraction, and classification stages.

### Convolutional feature extraction

The overall network architecture was composed of an input stage, a feature extraction stage with three convolutional layer blocks, and an output provision classification stage. The input stage received the preprocessed dataset. The feature extraction stage followed a triple ternary convolution block structure including convolution, pooling, and activation functions. The last classification stage consisted of a fully connected layer, dropout, and softmax operator. The final network outputs came from the softmax operation results.

(i) Input layer: In a CNN, a layer can be represented by the general expression

function $y = f(x)$where both the input x and output y are tensors. The network input is a Height(H) Width(W) Dimension(D) sized tensor data. In this study, the input was the time series accelerometer data where H = window size, W = single dimension, and D = number of sensor axes. As shown in Fig. 3.10, this input size H WD was equal to 6413.

(ii) Convolution layer: The convolution layer computed the input x convolution using filters f in order to detect local conjunctions of features from the previous layer and mapping their appearance to a feature map. However, in this study we considered two types of convolutional layers, one with uni-dimensional kernels and one with di-dimensional kernels as shown in Fig. 3.11 . 1D kernels are used to extract temporal dependency over the signal while CNN with 2D kernel is able to captures spatial dependency over sensors and axes as well as local dependency over time at the same time. This issue has been discussed in the publication [Sojeong Ha, 2015].

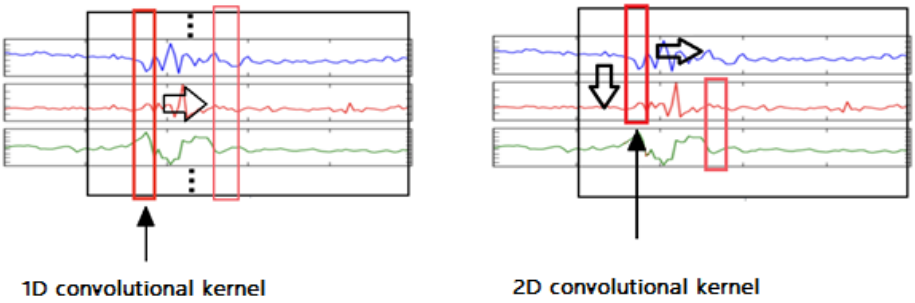



Figure 3.11: Convolutional neural network (CNN) kernels: (Left) convolutional layers with uni-dimensional kernels, (Right) convolutional layers with di-dimensional kernels .

(iii) Pooling layer: Pooling layers are used to identify the most important features and compresses layer sizes by subsampling. In this study, a max pooling method is used after a convolutional stage in order to reduce the computational requirements progressively through the network as well as minimizing the likelihood of overfitting.

(iv) Activation function: An activation function followed the max pooling step in the convolution block. A rectified linear unit (ReLU), was selected for this function to mitigate the vanishing gradient problem and reduce learning time [Glorot X, 2011]. Any negative output values remaining after the max pooling layer were set to zero.

(v) Repeated convolution blocks: Three convolution blocks were used, combining the steps of convolution, max pooling,and ReLU. Figure 3.10 shows the ternary convolution

block structure, the filter sizes of which were 32 for all the three blocks.

**Classification**

(i) Dropout layer: A dropout layer was imported to prevent overfitting, which may occur if a node in one layer excessively influences the subsequent layer. The dropout step dropped randomly selected nodes in fully connected layers.(Hinton et al., 2012;Srivastava et al., 2014). This step could be performed in every fully connected layer except the final one for output. In this study case, only a single dropout layer was inserted in the flatten fully connected layer using a rate of 0.2.

(ii) Fully connected layer: The ReLU results of each layer, as with all layer outputs, were connected to the weighted nodes of the next layer. This study featured 1024 empirically set nodes for the first fully connected layer and 256 empirically set nodes for the second one.

(iii) Softmax layer: The softmax function was applied across feature channels to consolidate network results. This normalization method was applied to make that the sum of output probability distributions equal to 1.

### 3.2.3 Evaluation Metrics

There exists a set of metrics to measure the activity recognition performance, such as accuracy, recall (also known as sensitivity), and area under the ROC curve (AUC).

First, there are 4 important terms to be considedred:
True Positives : The cases in which we predicted YES and the actual output was also YES. True Negatives : The cases in which we predicted NO and the actual output was NO. False Positives : The cases in which we predicted YES and the actual output was NO. False Negatives : The cases in which we predicted NO and the actual output was YES.

$$Accuracy = \frac{TruePositives + TrueNegatives}{Totalnumberofpredictionsmade} \tag{3.2}$$

**Classification accuracy**

Classification accuracy is the number of correct predictions made as a ratio of all predictions made. This is the most common evaluation metric for classification problems

**Area Under ROC Curve**

Area under ROC Curve (or AUC for short) is a performance metric for binary classification problems [Jason Brownlee, 2019]. The AUC represents a model's ability to discriminate between positive and negative classes. An area of 1.0 represents a model that made all predictions perfectly. An area of 0.5 represents a model as good as random. ROC can be broken down into sensitivity and specificity. A binary classification problem is really a trade-off between sensitivity and specificity.

**Sensitivity** is the true positive rate also called the recall. It is the number instances from the positive (first) class that actually predicted correctly.

$$Sensitivity = \frac{TruePositives}{TruePositives + FalseNegatives} \tag{3.3}$$

**Specificity** is also called the true negative rate. Is the number of instances from the negative class (second) class that were actually predicted correctly.

$$Specificity = \frac{FalsePositives}{FalsePositives + TrueNegatives} \tag{3.4}$$

Though, an extension of the ROC curve to multiclass forecast problems is explored in [WANDISHIN and MULLEN, 2008] where we could treat the multiclass as a binary classification problem and to consider each class vs all the other classes and calculate the operating points for each class and then average it out for the entire classifier.

In this work the accuracy F-measure and area on the curve (AUC) are used to evaluate our model.

### 3.2.4 Validation Protocols

Cross-validation is an fundamental step in classification tasks to separate the available data into training and testing sets. With this intention, many works used several cross-validation techniques namely k-fold, Leave-One-Subject-Out (LOSO), Hold-Out, Leave-One-Trial-Out (LOTO), Leave-One-Sample-Out, etc... K-fold (with k = 10) and Leave-One-Subject-Out are the most traditionally used protocols in the context of human activity recognition. Few works used Leave-One-Sample-Out because of high computational cost due to the high number of iterations.

### K-fold cross-validation

The procedure has a unique parameter called k which refers to the number of folds in which the data will be split into. As such, the procedure is often called cross-validation of k-folds. When a specific value is chosen, it can be used in the model reference, such as cross-validation multiplied by 10.

It is a popular method because of its advantages such as its simplicity to understand and its ability to return results in a less biased or less optimistic estimate of the model skill than other methods, such as a simple train/test split. Also the variance of the resulting estimate is reduced as k increases. However it limitation reside in the fact that it allow some of data of a specific subject to be used in both training and testing in the same iteration. Therefore, the resulted model will not be independent towards subjects.

The general procedure is as follows:

**Algorithm 2:** K-fold cross validation process

**Input :** All data

**Output:** Score of the model

1 Shuffle the dataset randomly.
2 Split the dataset into k groups
3 **for** *each unique group* **do**
4 | Take the group as a hold out or test data set
5 | Take the remaining groups as a training data set
6 | Fit a model on the training set and evaluate it on the test set
7 | Retain the evaluation score and discard the model
8 **end**
9 Summarize the skill of the model using the sample of model evaluation scores

### LOSO cross-validation

Leave-one-subject-out cross-validation (LOSO) is a particular case of K-fold cross-validation where a subject can be seen as a fold. Hence, the number of subjects determine the number of folds. It involves using the observations of one subject as the validation set and the remaining observations as the training set. Hence, with this technique we guaranty that we do not have subject bias.

### Nested cross-validation

Another used cross validation protocol is the Nested cross-validation. It is mainly used for the selection of the best set of hyperparameters and for error estimation (and assessment of generalization capacity) since if we use the test set to both select the values of the parameter and evaluate the model, we risk optimistically biasing our model evaluations. Many variants exist, but the principle is the same. First, an inner cross validation is used to tune the parameters and select the best model. Second, an outer cross validation is used to evaluate the model selected by the inner cross validation.

**Algorithm 3:** Nested cross validation process

**Input :** All data

**Output:** Score of the model

```
Divide the dataset into K cross-validation folds at random. for each fold
  k=1,2,..., K: outer loop for evaluation of the model with selected
  hyperparameter do
    Let test be fold k
    Let trainval be all the data except those in fold k
    Randomly split trainval into L folds for For each fold l=1,2,... L: inner
      loop for hyperparameter tuning do
        Let val be fold l
        Let train be all the data except those in test or val
        Train with each hyperparameter on train, and evaluate it on val. Keep
          track of the performance metrics
        For each hyperparameter setting, calculate the average metrics score over
          the L folds, and choose the best hyperparameter setting.
        Train a model with the best hyperparameter on trainval. Evaluate its
          performance on test and save the score for fold k.
    end
end
Calculate the mean score over all K folds, and report as the generalization error.
  Summarize the skill of the model using the sample of model evaluation scores
```

In this work, we used the k-fold and LOSO cross-validation protocols to validate our models. In fact, we are aiming to develop a novel model for HAR that meet our

fundamental required criteria which is its independence towards tsubjects. There for LOSO cross-validation protocol will be a adequate protocol in this case where a subject can be seen as a fold for testing in each iteration. The number of subjects will determine the number of iterations (folds). We have also used nested cross-validation to tune and evaluate our machine learning model.

# Chapter 4

# Result and Analysis

This chapter aims to present the results of the implemented work. We described both the hardware and software environment used in the development. Also we gave an overview of the obtained results.

## 4.1 Work environments

The work environments cover both the Hardware environment and the software environment.

### 4.1.1 Hardware environment

- **System Manufacturer:** LENOVO
- **Processor:** Intel(R) Core(TM) i5-7200U CPU @ 2.50GHz, 2701 MHz, 2 Core(s), 4 Logical Processor(s).
- **Installed Physical Memory (RAM):** 8.00 GB
- **GPU:** Nvidia Geforce 945MX, (Cuda anbled GPU with 384 cores 4.00 GB dedicated video memory)
- **OS Name:** Microsoft Windows 10 Enterprise

Essentially we have used the GPU to execute the the CNN algorithms since it is embedded with multiple cores dedicated for video and image processing and enables

faster execution than the CPU. In particular, the convolution computing is faster in GPUs thanks to the circuits implemented within it.

### 4.1.2 Software environment

Python is an interpreted, high-level, general-purpose programming language. Created by Guido van Rossum and first released in 1991, Python has a design philosophy that emphasizes code readability, notably using significant whitespace. It provides constructs that enable clear programming on both small and large scales.

MATLAB also provides great libraries and packages for specific tasks of image analysis. You can find great number of toolboxes for the given task. The main con of using MATLAB is that it is very slow(execution time is slow). It can't be used in deployment, but only for prototyping. Also it's not free to use, unlike python which is open.

Another great tool is R. It's an open source tool, free and made for statistical analysis. However Python is a great tool for the development of programs which perform data manipulation whereas R is a statistical software which works on a particular format of dataset. Python provides the various development tools which can be used to work with other systems.

Given these points, Python is mainly used for the developing of the classification system. Hence I used many python licensed libraries:

1. Pandas for data manipulation and analysis
2. Numpy for mathematical and logical operations on data arrays
3. Scikit-learn for building ML models
4. Keras for building DL models

## 4.2 Database

As a first step of this project, we loaded the data of the first 22 participants. Recall that each participant is asked to do, 6 times, a cycle containing the following physical activities: climb (C1) and go down the stairs (C2), walk (C3), run (C4), sit (C5 ), falls (C6) and lie down (Cl7). Classes C6 and C7 includes four types of falls and four types of lying respectively: left, right, back and front.

After performing the segmentation of the dataset (with the sliding window of 1 second for the CNN model and with peak detection for the SVM model) we obtained a total number of obtained equal to 30684 segement (distributed as shown in figure 4.1) and a total number of 12278 segments (distributed as shown in figure 4.2) respectively for the CNN model and the SVM model.

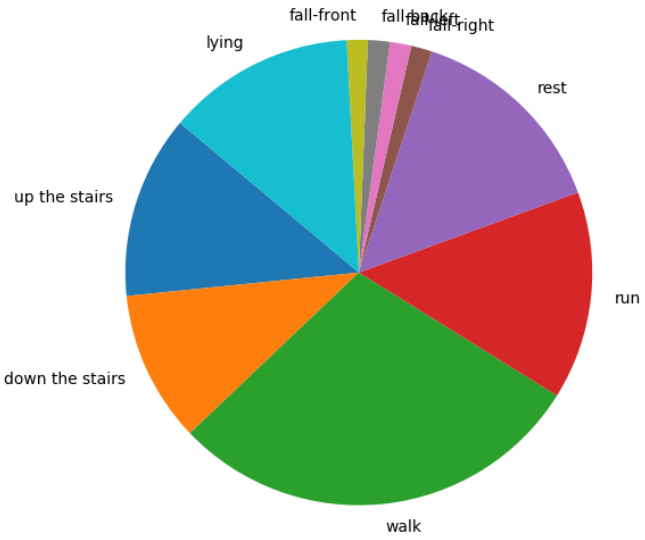



Figure 4.1: Database distribution for the CNN model.




Figure 4.2: Database distribution for the SVM model

At first sight we can see that the data are highly imbalanced. In that case, we propose two techniques. The first technique is one of the simplest ways to address the class imbalance. It simply provide a weight for each class which places more emphasis on the minority classes such that the end result is a classifier which can learn equally from all classes. In a gradient-based model, class weight is used scale the calculated loss for each observation by the appropriate value such that you place more significance on the losses associated with minority classes as described in [**?**] and [**?**].

The second technique is over and under sampling. Undersampling means we will select only some of the data from the majority class, only using as many examples as the minority class has. However, with this technique we will be wasting a huge amount of data (not recommended for deep learning algorithms ) since the number of samples in each class will be equal to the number of samples in the smallest one. It can discard potentially useful information. Also The sample chosen by under sampling may be a biased sample. And it will not be an accurate representative of the population. On the other hand, oversampling means that we will create copies of our minority class in order to have the same number of examples as the majority class has. The copies will be made such that the distribution of the minority class is maintained. However we might artificially reduce the variance of the dataset and increase the likelihood of overfitting since it replicates the minority class events.

Ultimately, we used the first technique to handle our imbalanced dataset. We specified class weights so they are inversely proportional to the size of the class.

$$Class_weight(C_i) = \frac{Total\ number\ of\ samples}{Number\ of\ classes \times Number\ of\ samples\ of\ class(C_i)} \tag{4.1}$$

## 4.3 Experimental Results

The developed classification systems (i.e. SVM, CNN(1D kernel) and CNN(2D kernel)) were evaluated in terms of good classification rates. For the SVM the characteristic vectors are formed by the concatenation of all the characteristics previously described (the average value of the signal, the amplitude, the minimum, the maximum and the range as well as the asymmetry) computed around the detected peaks. Discriminant characteristics were selected from all of the characteristics using the Relief-F algorithm.

The tests were done according to the process of Leave-one-subject-out cross vali-

dation and 10-fold cross validation for each of the three developped models. Table 4.1 summarizes the results obtained for the three classifiers developed.We note that the classification rates are globally higher with the CNN(2D).

| | SVM | CNN(1D kernel) | CNN(2D kernel) |
|---|---|---|---|
| 10-fold cross vladidation | 96% | 93% | 94% |
| Leave One Subject Out cross validation | 72% | 76% | 84% |
| Complexity | Low | High | Very high |

Table 4.1: Classification rate of the different classifiers

This result is predictable since the CNN algorithm will extract high-level features in an incremental manner. It will find the most proprieties that best describe the signal since with the traditional technique for features engineering, the calculated features may not describe very well the signal in correlation with the corresponding classes. Also, CNN with 2D kernels will best extract more relevant features than the one with 1D kernel since it takes advantage of the dependency within the three axis of the acceleration signal.

For a better investigation of the results, we have represented the ROC curves of the CNN(1D) and CNN(2D). Recall that ROC is a probability curve and AUC represents degree or measure of separability. It tells how much model is capable of distinguishing between classes. Higher the AUC, better the model is at predicting 0s as 0s and 1s as 1s. As we can see, it our two model based on CNN have an an AUC superior to 0.97 for all classes. Both are excellent models with AUC near to the 1 which means they have good measure of separability.

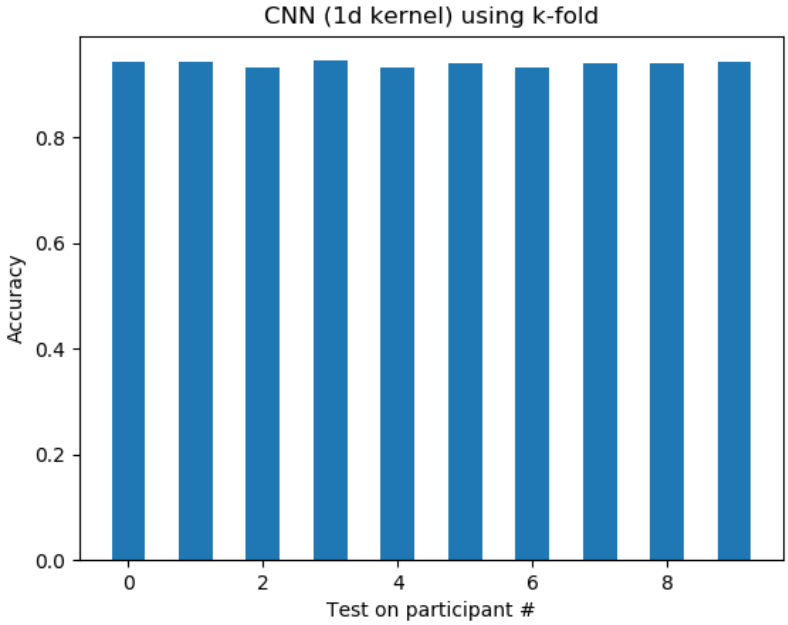



Figure 4.3: CNN (1D kernel) results with k-fold

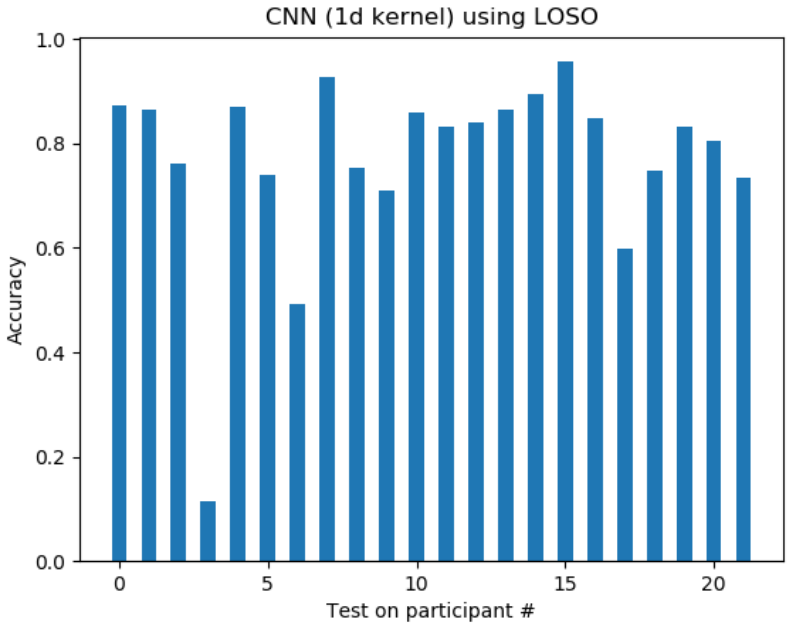



Figure 4.4: CNN (1D kernel) results with LOSO

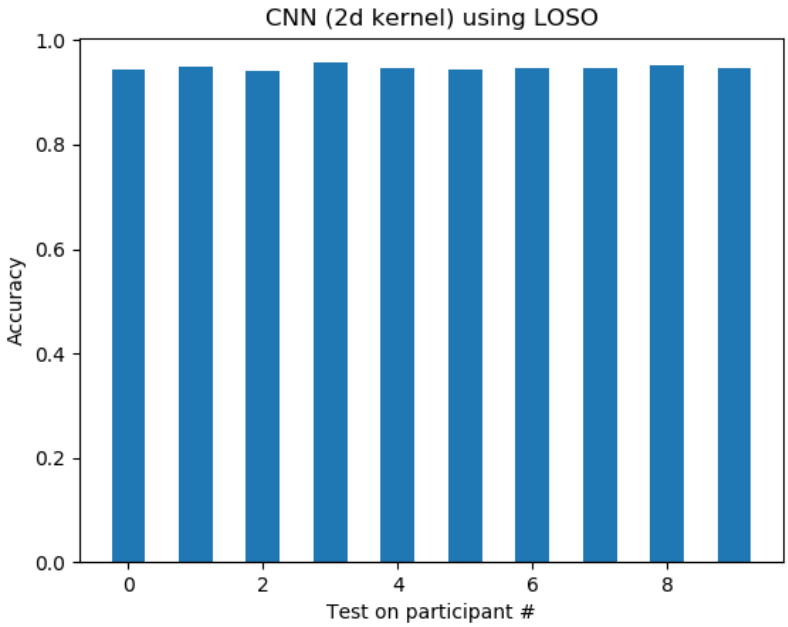



Figure 4.5: CNN (2D kernel) results with 10-fold cv

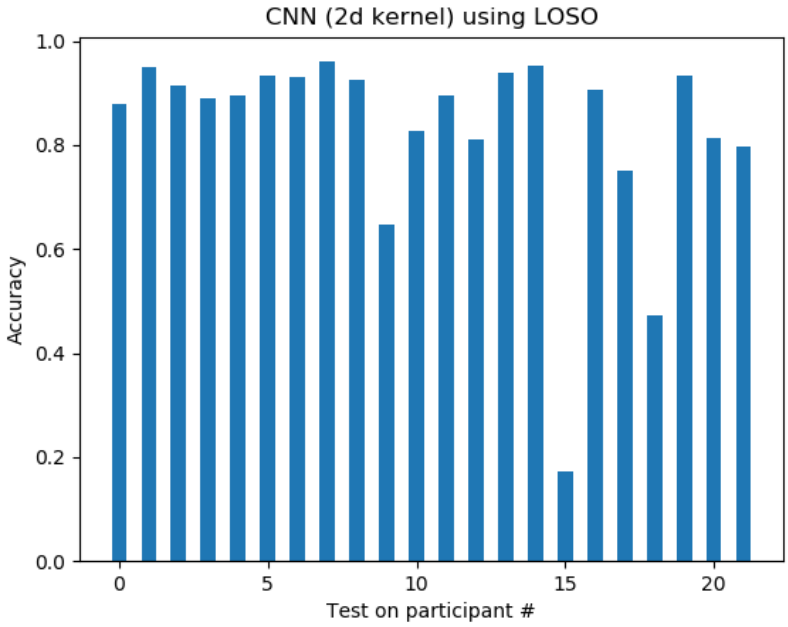



Figure 4.6: CNN (1D kernel) results with LOSO

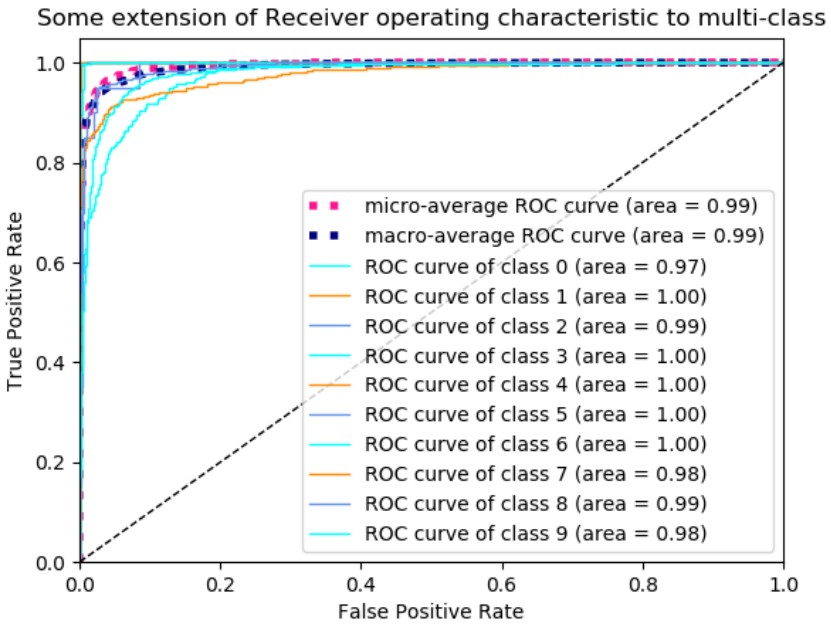



Figure 4.7: ROC curve of the CNN (1D kernel) results with LOSO

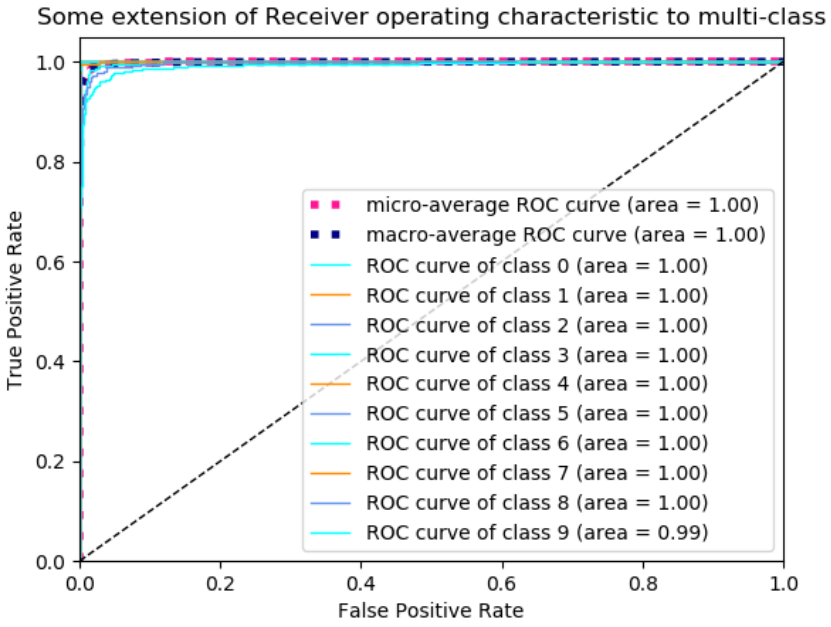



Figure 4.8: ROC curve of the CNN (2D kernel) results with LOSO

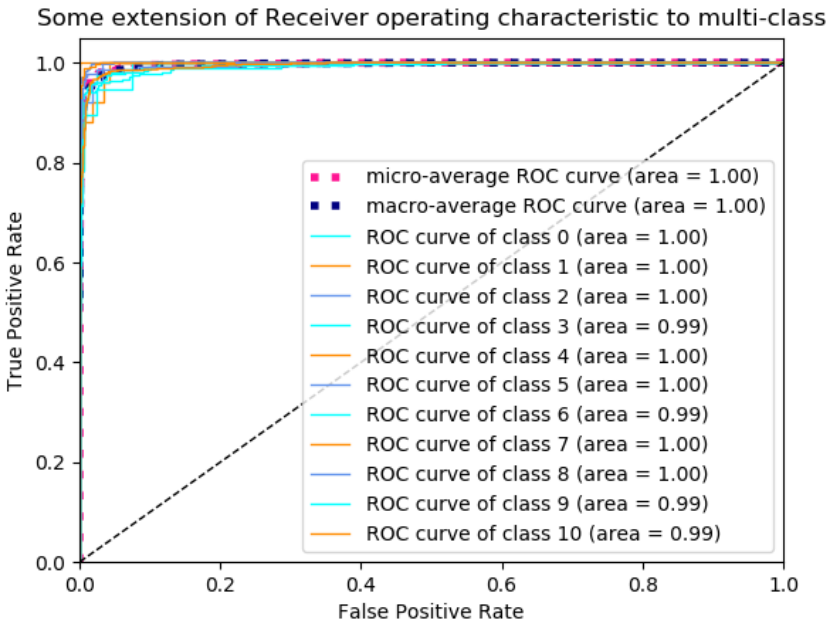



Figure 4.9: ROC curve of the CNN (1D kernel) results with 10-fold cv

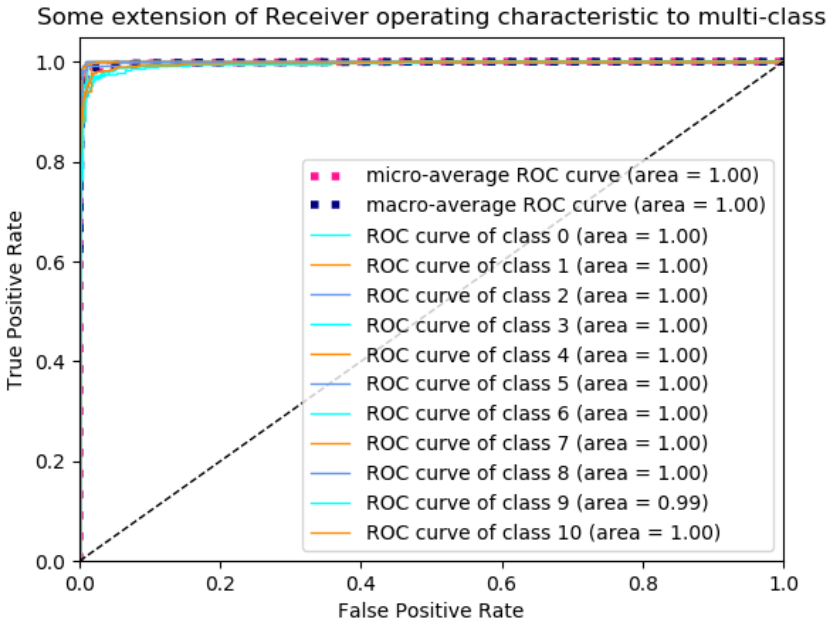



Figure 4.10: ROC curve of the CNN (2D kernel) results with 10-fold cv

# Chapter 5

# Conclusion and Future Work

In this study, the classification of human physical activity was performed. This study involved algorithm development, participant recruitment, Ethic approval, and data collection. The developed algorithm distinguished physical activities into 10 different classes.

Data collection took place at CRCHUM (Research Center of the University Hospital of Montreal) with the participation of 44 people. Each participant is asked to do, six times, a cycle containing in sequence the following 7 physical activities: climbing (C1) and descending the stairs (C2), walking (C3), running (C4), sitting down ( C5), falls (C6) and lying (C7). Classes C6 and C7 includes four types of falls: left, right, back and front.

We subsequently developed three classification models were used, namely, the SVM as a conventional machine learning technique, and the two variant of the CNN, the first CNN is defined with uni-dimensional (1D) kernels used to extract local dependency over the time. The second CNN is defined with di-dimensional (2D) kernels used to extract local dependency over the time as well as the spatial dependency over the the different axis of the acceleration signal.

The results obtained show that the CNN with 2D kenels allowed the best classification rates (94 with k-fold cross validation protocol and 84 perecent with the LOSO cross validation protocol) compared to CNN with 1D kernels.

This study represents some limitations such as the the number of subjects was restricted to twenty-two participants which is limited for the validation of the results. To do this, we suggest increasing the size of the database by adding the other twenty-two subjects data to test and validate the different classifiers used in this study.

Other lines of research might interest researchers in the field of automatic classi-

fication by using other signal such as gyroscope signal. Under those circumstances, we aim to use physiological data in the classification such electrocardiogram signal and the respiratory signal provided by the hexoskin textile.

# Bibliography

[Artur Jordao, 2018] Artur Jordao, Ricardo Kloss, W. R. S. (2018). Latent hypernet: Exploring the layers of convolutional neural networks.

[Banos O, 2014] Banos O, Galvez JM, D. M. P. H. R. I. (2014). Window size impact in human activity recognition. *Sensors (Basel)*, 14(4):6474—6499.

[Bruno et al., 2015] Bruno, B., Mastrogiovanni, F., and Sgorbissa, A. (2015). Wearable inertial sensors: Applications, challenges, and public test benches. *Robotics Automation Magazine, IEEE*, 22:116–124.

[C. Chen and Kehtarnavaz, 2015] C. Chen, R. J. and Kehtarnavaz, N. (2015). Utd-mhad: A multimodal dataset for human action recognition utilizing a depth camera and a wearable inertial sensor. *IEEE*.

[Diego Castro, 2015] Diego Castro, William Coral, C. R. J. C. J. C. (2015). Wearable-based human activity recognition using and iot approach. *Sensor and Actuator Networks*, 8.

[Donges, 2019] Donges, N. (2019). Pros and Cons of Neural Networks. https://towardsdatascience.com/hype-disadvantages-of-neural-networks-6af04904ba5b. [Online; accessed 12-June-2019].

[Fournier and Roy, 2011] Fournier, P. A. and Roy, J. F. (2011). A smart shirt for monitoring heart, breathing and other activities.

[Glorot X, 2011] Glorot X, Bordes A, B. Y. (2011). Deep sparse rectifier neural networks. in: Proceed-ings of the fourteenth international conference on artificial intelligence and statistics. pages 315—-323.

[healthcare, 2019a] healthcare, A. P. (2019a). AXA PPP healthcare. https://www.axappphealthcare.co.uk/. [Online; accessed 29-Mai-2019].

[healthcare, 2019b] healthcare, A. P. (2019b). wisdm dataset. http://www.cis.fordham.edu/wisdm/dataset.php. [Online; accessed 30-Mai-2019].

[I. Kononenko, 1997] I. Kononenko, E. Simec, M. R. S. (1997). Overcoming the myopia of inductive learning algorithms with relief. *Applied Intelligence*, 7:39–55.

[Jang et al., 2018] Jang, Y., Kim, S., Kim, K., and Lee, D. (2018). Deep learning-based classification with improved time resolution for physical activities of children. *PeerJ*, 6:e5764.

[Jason Brownlee, 2019] Jason Brownlee, P. (2019). Assessing and Comparing Classifier Performance with ROC Curves. https://machinelearningmastery.com/assessing-comparing-classifier-performance-roc-curves-2/. [Online; accessed 29-Mai-2019].

[Jessica Sena, 2018] Jessica Sena, Jesimon Barreto Santos, W. R. S. (2018). Wearable inertial sensors: Applications, challenges, and public test benches.

[Jian Cui, 2013] Jian Cui, B. X. (2013). Cost-effective activity recognition on mobile devices. *ACM*, 7:90–96.

[Krause et al., 2005] Krause, A., Ihmig, M., Rankin, E., Leong, D., Gupta, S., Siewiorek, D., Smailagic, A., Deisher, M., and Sengupta, U. (2005). Trading off prediction accuracy and power consumption for context-aware wearable computing. *2012 16th International Symposium on Wearable Computers*, 0:20–26.

[M. Robnik-Sikonja, 2003] M. Robnik-Sikonja, I. K. (2003). Theoretical and empirical analysis of relieff and rrelieff. *Machine learning*, 53:23–69.

[N. Mezghani, 2017] N. Mezghani, Y. Ouakrim, M. R. I. R. Y. B. A. (2017). Context aware adaptable approach for fall detection bases on smart textile. *IEEE*, 5:473–476.

[Nour Cherif, 2018a] Nour Cherif, Youssef Ouakrim, A. B.-B. N. M. (2018a). *Physical activity classification using a smart textile.* ©2018 IEEE.

[Nour Cherif, 2018b] Nour Cherif, Youssef Ouakrim, A. B.-B. N. M. (2018b). Physical activity classification using a smart textile. ©*2018 IEEE*, 4:175–178.

[O. Banos and Villalonga, 2014] O. Banos, R. Garcia, J. A. H. T. M. D. H. P. I. R. R. A. S. and Villalonga, C. (2014). A novel framework for agile development of mobile health applications. *IWAAL*, 8:91–98.

[Pyle, 1999] Pyle, D. (1999). Data preparation for data mining. *Diane D. Ed.; Morgan Kaufmann Inc.*

[Reiss and Stricker, 2012] Reiss, A. and Stricker, D. (2012). Introducing a new benchmarked dataset for activity monitoring. *IEEE*, 8:91–98.

[Shuochao Yao, 2015] Shuochao Yao, Shaohan Hu, Y. Z. A. Z. T. A. (2015). Deepsense: A unied deep learning framework for time-series mobile sensing data processing. *ACM*, 8.

[Sojeong Ha, 2016] Sojeong Ha, S. C. (2016). Convolutional neural networks for human activity recognition using multiple accelerometer and gyroscope sensors. *IEEE*, 8:381–388.

[Sojeong Ha, 2015] Sojeong Ha, Jeong-Min Yun, S. C. (2015). Multi-modal convolutional neural networks for activity recognition. *IEEE*, 6:3017–3022.

[WANDISHIN and MULLEN, 2008] WANDISHIN, M. S. and MULLEN, S. J. (2008). Multiclass roc analysis. 24:530–547.

[Wenchao Jiang, 2015] Wenchao Jiang, W. Y. (2015). Human activity recognition using wearable sensors by deep convolutional neural networks. *IEEE*, 4.

[Yuqing Chen, 2015] Yuqing Chen, Y. X. (2015). A deep learning approach to human activity recognition based on single accelerometer. *IEEE*, 5:1488–1492.